



Report No(报告编号): HTT202510122CH

RoHS Test Report(ROHS 测试报告)

Application information(客户信息):

Applicant name(客户名):	Xinxiang Hongli Supply Source Technology Co.,Ltd. (新乡市弘力电源科技有限公司)
Address(地址):	West Section of Xinglong Road, Xinxiang Economic Development Zone, Xinxiang County, Xinxiang City, Henan, P.R. China (中国河南省新乡市新乡县新乡经济开发区兴隆路西段)
Manufacturer(制造商):	Xinxiang Hongli Supply Source Technology Co.,Ltd. (新乡市弘力电源科技有限公司)
Address(地址):	West Section of Xinglong Road, Xinxiang Economic Development Zone, Xinxiang County, Xinxiang City, Henan, P.R. China (中国河南省新乡市新乡县新乡经济开发区兴隆路西段)

Sample information(样品信息):

Sample Name(样品名):	Cylindrical lithium-ion rechargeable cell (圆柱形可充电锂离子电芯)
Sample Model(样品型号):	IMR18650-1200mAh, IMR18650-1800mAh, IMR14500-300mAh, MR14500-400mAh, IMR14500-500mAh, IMR14500-600mAh, IMR18650-600mAh, IMR18650-700mAh, IMR18650-800mAh, IMR18650-1050mAh, IMR18650-1200mAh, IMR18650-1500mAh, IMR18650-1800mAh, IMR18650-1300mAh, IMR18650-2000mAh
Trade mark(商标):	N/A (无)
Sample Received Date (样品接收日期):	Oct. 09, 2025 (2025.10.09)
Testing Period (测试周期):	Oct. 09, 2025 ~ Oct. 14, 2025 (2025.10.09 - 2025.10.14)
Test Requested (测试项目):	As specified by client, to determine the Lead, Cadmium, Mercury, Hexavalent Chromium, PBBs, PBDEs, The Phthalate(DEHP, BBP, DBP & DIBP) content in the submitted sample compliance with RoHS directive 2011/65/EU Annex II amending Annex (EU)2015/863 and amending Annex (EU)2017/2102.
Test Method(测试方法):	Please refer to next page.
Test Results(测试结果):	Please refer to next page(s).
Conclusion(结论):	PASS

Completed by(完成人):

Louis. Liu

Reviewed by(审核人):

Leo Zhang

Approved by(批准人):

Kevin Yang

Technical Manager(技术经理)



Shenzhen HTT Technology Co., Ltd.
深圳市环通检测技术有限公司



Report No(报告编号). HTT202510122CH

RoHS Test Report(ROHS 测试报告)

Test Method(s)(测试方法):

As requested by the client, Reference to IEC 62321-3-1:2013 Procedures for the screening analysis of Lead (Pb), Cadmium (Cd), Mercury (Hg), total Chromium (Cr), and Bromine (Br) by XRF. If the screening analysis results exceed the screening limits of IEC 62321-3-1:2013 Annex A, use the chemical methods for testing.

(应客户要求, 参考 IEC 62321-3-1:2013 XRF 筛查分析铅 (Pb)、镉 (Cd)、汞 (Hg)、总铬 (Cr) 和溴 (Br) 的程序。如果筛选分析结果超过 IEC 62321-3-1:2013 Annex A 的筛选限值, 则使用化学方法进行测试。)

Table1 IEC 62321-3-1:2013 Annex A screening limits of XRF (mg/kg)

(表 1 IEC 62321-3-1:2013 Annex A XRF 筛查限值 (mg/kg))

Element (元素)	Polymer (聚合物)	Metal (金属)	Composite Materials (复合材料)
Cd	$BL \leq (70-3\sigma) < X < (130+3\sigma) \leq OL$	$BL \leq (70-3\sigma) < X < (130+3\sigma) \leq OL$	$LOD < X < (150+3\sigma) \leq OL$
Pb	$BL \leq (700-3\sigma) < X < (1300+3\sigma) \leq OL$	$BL \leq (700-3\sigma) < X < (1300+3\sigma) \leq OL$	$BL \leq (500-3\sigma) < X < (1500+3\sigma) \leq OL$
Hg	$BL \leq (700-3\sigma) < X < (1300+3\sigma) \leq OL$	$BL \leq (700-3\sigma) < X < (1300+3\sigma) \leq OL$	$BL \leq (500-3\sigma) < X < (1500+3\sigma) \leq OL$
Br	$BL \leq (300-3\sigma) < X$	Not applicable	$BL \leq (250-3\sigma) < X$
Cr	$BL \leq (700-3\sigma) < X$	$BL \leq (700-3\sigma) < X$	$BL \leq (500-3\sigma) < X$

BL = Less than screening limits of XRF(低于 XRF 的筛查限值)

OL = More than screening limits of XRF(高于 XRF 的筛查限值)

X = The results of screening analysis by XRF are within this range, requiring further chemical testing.(此范围内的筛查结果需进一步的化学测试确认)

LOD = Limit of Detection(检出限)



Report No(报告编号). HTT202510122CH

RoHS Test Report(ROHS 测试报告)

Table2 testing methods & Equipments**(表 2 测试方法和设备)**

Testing Item(测试项目)	Testing Method(测试方法)	Equipment(仪器设备)
Screening analysis by XRF(XRF 筛查分析):		
Lead (Pb)(铅), Cadmium (Cd)(镉), Mercury (Hg)(汞), Chromium (Cr)(铬), Bromine (Br)(溴)	IEC 62321-3-1:2013	ED-XRF
Chemical testing(化学测试)		
Lead (Pb)(铅)	IEC 62321-5:2013	ICP-OES
Cadmium (Cd)(镉)	IEC 62321-5:2013	ICP-OES
Mercury(Hg)(汞)	IEC 62321-4:2013+A1:2017	ICP-OES
Hexavalent chromium (Cr(VI)) for plastic (聚合物中的六价铬)	IEC 62321-7-2:2017	UV-VIS
Hexavalent chromium (Cr(VI)) for coating on metals(金属涂层中的六价铬)	IEC 62321-7-1:2015	UV-VIS
PBBs(多溴联苯)	IEC 62321-6:2015	GC-MS
PBDEs(多溴联苯醚)	IEC 62321-6:2015	GC-MS
DBP(邻苯二甲酸二丁酯)	IEC 62321-8:2017	GC-MS
BBP(邻苯二甲酸丁基苄酯)	IEC 62321-8:2017	GC-MS
DEHP(邻苯二甲酸二(2-乙基己基)酯)	IEC 62321-8:2017	GC-MS
DIBP(邻苯二甲酸二异丁酯)	IEC 62321-8:2017	GC-MS



Report No(报告编号). HTT202510122CH

RoHS Test Report(ROHS 测试报告)

1. Method Detection Limit (方法检出限)(MDL):

For XRF screening analysis(XRF 筛查分析) (mg/kg)

Item(项目)	Pb	Cd	Hg	Br	Cr
Polymer(聚合物)	20	20	20	20	20
Other materials(其他材质)	50	50	50	50	50

For chemical testing(化学测试) (mg/kg)

Item(项目)	Pb	Cd	Hg	PBBs	PBDEs	DEHP	BBP	DBP	DIBP	Cr(VI)
General materials (一般材料)	10	10	10	10	10	50	50	50	50	See remark (见备注) (2)

Remark(备注):

(1) PBBs and PBDEs method detection limit only for one substance(PBBs 和 PBDEs 方法检出限仅适用于其中一种物质);

(2) MDL for polymer and Composites is 8 mg/kg, MDL for coating on metals is 0.10 µg/cm². (聚合物和复合材料的 MDL 为 8 mg/kg, 金属涂层的 MDL 为 0.10 µg/cm²).

2. RoHS Requirement(RoHS 限值要求) (mg/kg):

Restricted Substances (受限制)	Cd	Pb	Hg	Cr(VI)	PBBs	PBDEs	DEHP	BBP	DBP	DIBP
RoHS limit	100	1000	1000	1000	1000	1000	1000	1000	1000	1000



Report No(报告编号). HTT202510122CH

RoHS Test Report(ROHS 测试报告)

3. Test Result(s)(测试结果):

Part No. (材料号)	Part Description (材料描述)	Restricted Substances (受限制物质)	Results(结果) (mg/kg)		Conclusion (结论) (P/F)
			EDXRF (XRF 扫描)	Chemical testing(化学测试)	
1	Purple sticker (紫色贴纸)	Cd	BL	/	P
		Pb	BL	/	P
		Hg	BL	/	P
		Cr(VI)	BL	/	P
		PBBs	BL	/	P
		PBDEs	BL	/	P
		DEHP	/	N.D.	P
		DBP	/	N.D.	P
		BBP	/	N.D.	P
		DIBP	/	N.D.	P
2	White plastic (白色塑胶)	Cd	BL	/	P
		Pb	BL	/	P
		Hg	BL	/	P
		Cr(VI)	BL	/	P
		PBBs	BL	/	P
		PBDEs	BL	/	P
		DEHP	/	N.D.	P
		DBP	/	N.D.	P
		BBP	/	N.D.	P
		DIBP	/	N.D.	P
3	Silvery metal (银色金属)	Cd	BL	/	P
		Pb	BL	/	P
		Hg	BL	/	P
		Cr(VI)	BL	/	P
		PBBs	/	/	P
		PBDEs	/	/	P
		DEHP	/	/	P
		DBP	/	/	P
		BBP	/	/	P
		DIBP	/	/	P
4	Silvery meta (银色金属)	Cd	D	N.D.	P
		Pb	BL	/	P
		Hg	BL	/	P
		Cr(VI)	BL	/	P
		PBBs	/	/	P
		PBDEs	/	/	P
		DEHP	/	/	P
		DBP	/	/	P
		BBP	/	/	P
		DIBP	/	/	P



Report No(报告编号). HTT202510122CH

RoHS Test Report(ROHS 测试报告)

Part No. (材料号)	Part Description (材料描述)	Restricted Substances (受限制物质)	Results(结果) (mg/kg)		Conclusion (结论) (P/F)
			EDXRF (XRF 扫描)	Chemical testing(化学测试)	
5	White plastic (白色塑胶)	Cd	BL	/	P
		Pb	BL	/	P
		Hg	BL	/	P
		Cr(VI)	BL	/	P
		PBBs	BL	/	P
		PBDEs	BL	/	P
		DEHP	/	N.D.	P
		DBP	/	N.D.	P
		BBP	/	N.D.	P
		DIBP	/	N.D.	P
6	Silvery metal (银色金属)	Cd	BL	/	P
		Pb	BL	/	P
		Hg	BL	/	P
		Cr(VI)	BL	/	P
		PBBs	/	/	P
		PBDEs	/	/	P
		DEHP	/	/	P
		DBP	/	/	P
		BBP	/	/	P
		DIBP	/	/	P
7	Copper aluminum foil (铜色铝箔纸)	Cd	BL	/	P
		Pb	BL	/	P
		Hg	BL	/	P
		Cr(VI)	BL	/	P
		PBBs	BL	/	P
		PBDEs	BL	/	P
		DEHP	/	N.D.	P
		DBP	/	N.D.	P
		BBP	/	N.D.	P
		DIBP	/	N.D.	P
8	White plastic (白色塑胶)	Cd	BL	/	P
		Pb	BL	/	P
		Hg	BL	/	P
		Cr(VI)	BL	/	P
		PBBs	BL	/	P
		PBDEs	BL	/	P
		DEHP	/	N.D.	P
		DBP	/	N.D.	P
		BBP	/	N.D.	P
		DIBP	/	N.D.	P



Report No(报告编号). HTT202510122CH

RoHS Test Report(ROHS 测试报告)

Part No. (材料号)	Part Description (材料描述)	Restricted Substances (受限制物质)	Results(结果) (mg/kg)		Conclusion (结论) (P/F)
			EDXRF (XRF 扫描)	Chemical testing(化学测试)	
9	Black powder (黑色粉末)	Cd	BL	/	P
		Pb	BL	/	P
		Hg	BL	/	P
		Cr(VI)	BL	/	P
		PBBs	BL	/	P
		PBDEs	BL	/	P
		DEHP	/	N.D.	P
		DBP	/	N.D.	P
		BBP	/	N.D.	P
		DIBP	/	N.D.	P
10	Yellow sticker (黄色贴纸)	Cd	BL	/	P
		Pb	BL	/	P
		Hg	BL	/	P
		Cr(VI)	BL	/	P
		PBBs	BL	/	P
		PBDEs	BL	/	P
		DEHP	/	N.D.	P
		DBP	/	N.D.	P
		BBP	/	N.D.	P
		DIBP	/	N.D.	P

Note(注释):

mg/kg = milligram per kilogram(毫克每千克) = ppm

N.D. = Not detected(未检出) (<MDL)

BL=Less than screening limits of XRF(低于 XRF 筛查的检出限)

D = DETECTED(不确定) = Inconclusive(待确认)

P = PASS(合格)

F = FAIL(不合格)

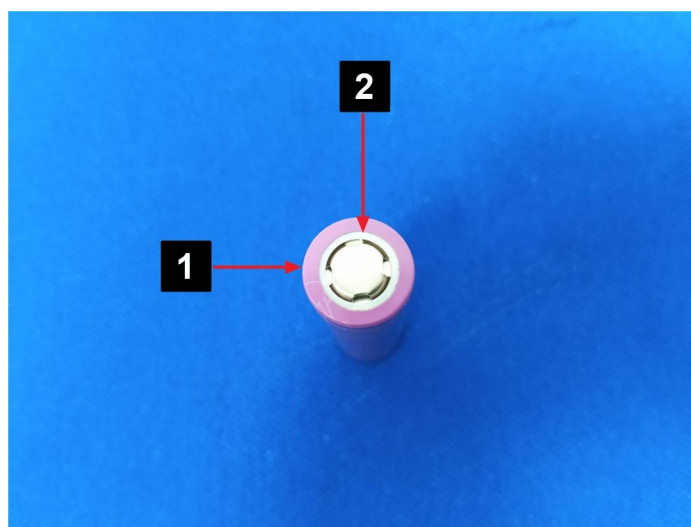
/ = Not available(不适用)



Report No(报告编号). HTT202510122CH

RoHS Test Report(ROHS 测试报告)

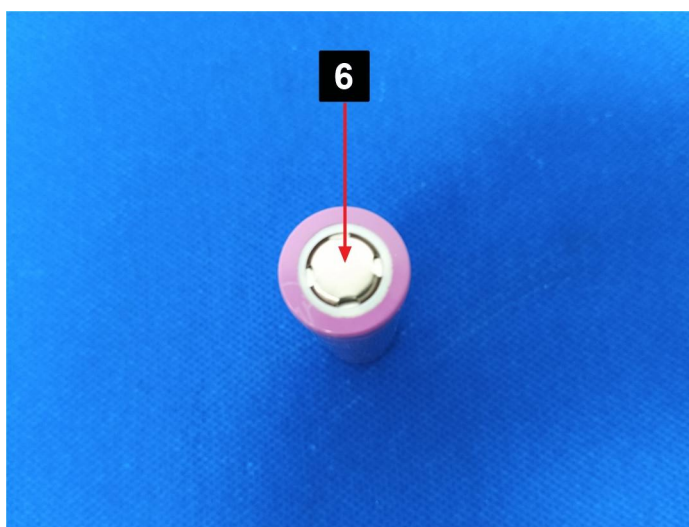
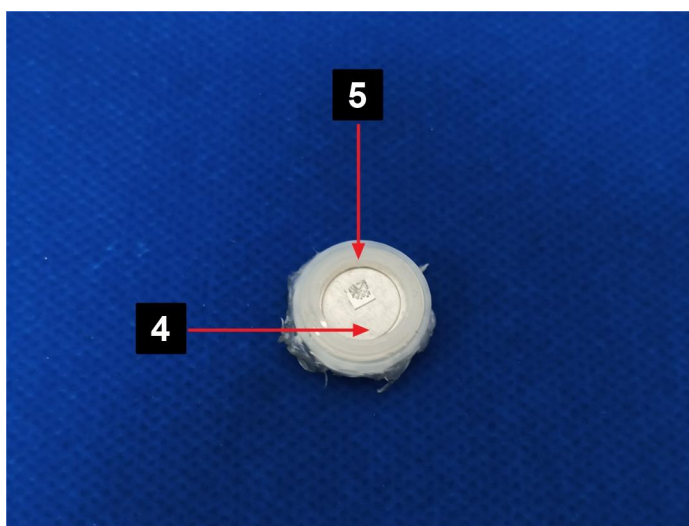
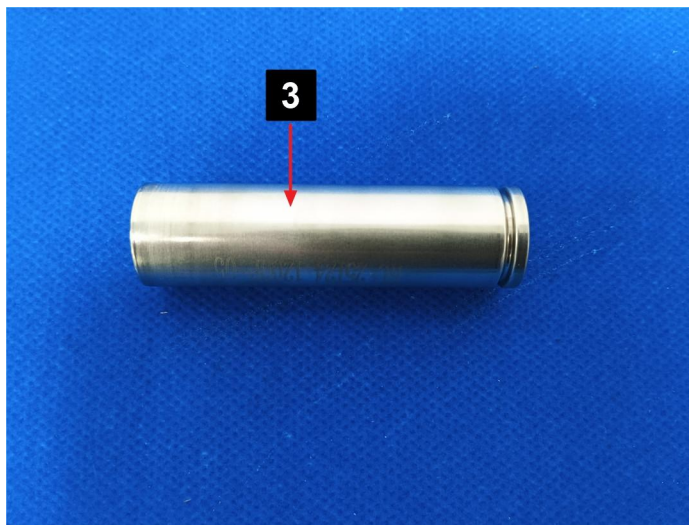
Photo(s) of the sample(s)(样品图片)





Report No(报告编号). HTT202510122CH

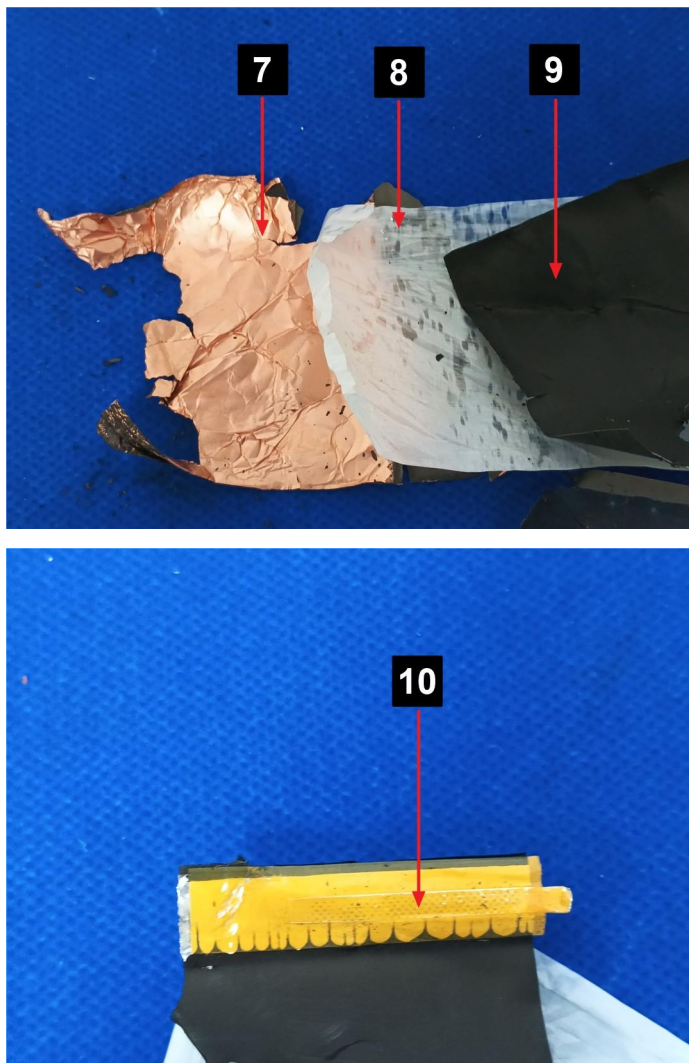
RoHS Test Report(ROHS 测试报告)





Report No(报告编号). HTT202510122CH

RoHS Test Report(ROHS 测试报告)



This test document cannot be reproduced in any way, except in full context, without the prior approval in writing of the laboratory. The result of this test report is based on our received samples only.

*** End of report ***

未经事先批准，不得以任何方式复制本测试文件，除非有完整的上下文
实验室的书面记录。本测试报告的结果仅基于我们收到的样品。

*** 报告结束 ***